Did Mandatory Clawbacks Curb CEO Risk-Taking? Difference-in-Differences Evidence from SEC Rule 10D-1

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Abstract

Voluntary clawback provisions are associated with *less* corporate risk-taking (Liu et al., 2020; Babenko et al., 2023). We ask whether the *mandatory* clawback regime created by SEC Rule 10D-1 (adopted October 2022; listing standards effective October 2023) produced the same effect. Because a compliant policy was required of essentially every listed issuer, a clean control group needs firms for whom the mandate did not bind. We exploit the fact that voluntary clawbacks were already near-universal at large firms but rare at small ones by 2013 (Babenko et al., 2023), and run a difference-in-differences comparing forced adopters (S&P SmallCap 600) with already-covered controls (S&P

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500), using 753 firms and 8,222 firm-years (2015–2025) with firm and year fixed effects and firm-clustered standard errors. **We find no robust evidence that the mandate changed CEO risk-taking.** The direct investment/M&A channels emphasized by the prior literature are flat: the post×treated coefficient is insignificant for investment/assets (+0.003, $t = 0.81$), capex (-0.001, $t = -1.30$), R&D (+0.002, $t = 1.33$), and acquisitions (+0.002, $t = 0.81$), as are leverage and cash. Stock-return volatility *does* fall for forced adopters (idiosyncratic vol. -0.027, $t = -5.01$), but this is not a clawback effect: parallel trends are rejected (pre-trend Wald $p < 0.001$), the event study shows no discontinuity at 2023, size-matching the treated and control firms attenuates it (to -0.021, $t = -2.37$), and size-based placebos that carry no clawback content reproduce it (S&P 400 vs. 500: -0.018, $t = -3.59$). Two additions push past the usual "we found nothing" paper. First, *randomization inference* (500 permutations of treatment) gives the investment null a design-based $p = 0.43$ while confirming the volatility coefficient is a systematic size phenomenon, not noise. Second, a *minimum-detectable-effect* analysis shows the null is *informative*: at 80% power we could have detected risk-taking changes of 5–16% of the outcome mean—smaller than the effects the voluntary-adoption literature reports—so the absence of an effect is evidence, not low power. Whether a true effect is hidden by our index-membership proxy for ex-ante clawback status is the question our hand-collection of 10D-1 provision strength is designed to answer.

JEL: G32, G34, G38, K22, M12. **Keywords:** clawback, executive compensation, risk-taking, SEC Rule 10D-1, difference-in-differences, randomization inference, statistical power.

1 Introduction

When a firm can *claw back* incentive pay after the fact, its executives bear more of the downside of risky bets, and standard contracting theory predicts they will take fewer of them (Coles et al., 2006). Empirically, firms that *voluntarily* adopted clawback provisions during 2005–2014 subsequently reduced risk-taking along investment and capital-structure margins (Liu et al., 2020), and Babenko et al. (2023) show that voluntary adoption causes measured firm risk to fall through more conservative investment and financial policies. In October 2022 the SEC finalized Rule 10D-1 under Dodd-Frank Section 954, and by late 2023 the national exchanges required *every* listed issuer to adopt a compliant clawback policy (SEC, 2022). This paper asks the natural follow-on question: did the *mandate* reproduce the risk-reduction that *voluntary* adoption delivered?

The central identification problem is that the mandate is close to universal, so a simple before/after comparison confounds the rule with everything else that happened to public firms in 2023–2025. We build a control group from the one dimension along which the mandate plausibly did *not* bind: firms that already had a voluntary clawback. Direct firm-level data on pre-existing voluntary policies is not available on our data platform, so we use the robust empirical regularity that voluntary clawbacks were near-universal among large firms but rare among small ones by 2013 (Babenko et al., 2023; Chen et al., 2015). We therefore treat the S&P SmallCap 600 as *forced adopters* (ex-ante mostly without a clawback) and the S&P 500 as *already-covered controls*, fix group membership at the pre-rule baseline (June 2022), and estimate a difference-in-differences with firm and year fixed effects. This is an intent-to-treat proxy; its validity, and the path to a sharper test, is the subject of the companion hand-collection task.

Our result is an honest null on the question that matters. Across the risk-taking outcomes the prior literature emphasizes, the mandate leaves no measurable footprint. In our preferred specification (firm + year fixed effects, controls, firm-clustered SE), the post$\times$treated coefficient is economically tiny and statistically insignificant for total investment/assets (+0.003, $t = 0.81$), capital expenditure (−0.001, $t = -1.30$), R&D (+0.002, $t = 1.33$), acquisitions (+0.002, $t = 0.81$), book leverage (+0.003, $t = 0.40$), and cash holdings (+0.008, $t = 1.49$) (Table 2). The one outcome that moves is stock-return volatility: idiosyncratic volatility falls by 0.027 for forced adopters ($t = -5.01$), and total volatility by 0.015 ($t = -2.60$).

We then show that this volatility result is *not* a clawback effect, and we take the space to do so because the temptation to report it as one is exactly the kind of false positive this literature must guard against. First, parallel trends are decisively rejected: a joint test of the pre-2022 dynamic-DiD coefficients gives Wald $p < 0.001$, and every pre-period coefficient is *positive* (Table 2, Figure 1)—forced adopters were structurally *more* volatile before the rule, not on a parallel path. Second, the event study shows no downward break at the 2023 effective date; the negative average "effect" is mechanically produced by the unusually large treated–control volatility gap during the 2020–21 COVID period mean-reverting afterward. Third, when we *size-match* treated and control firms on their pre-rule size—collapsing the treated–control size gap from 2.54 log points to 0.23—the volatility coefficient attenuates to −0.021 ($t = -2.37$), while the real-risk margins remain unsupportive of risk-reduction. Fourth, and decisively, size-based placebos that contain *no* differential clawback treatment produce the *same* coefficient: S&P MidCap 400 versus S&P 500 gives −0.018 ($t = -3.59$), and a below-median-size split gives −0.018 ($t = -4.02$) (Table 4). The volatility pattern is a small-firm × post-COVID phenomenon, not a response to Rule 10D-1.

Two further steps distinguish this from a paper that simply reports a coefficient near zero and stops. First, we subject the design to *randomization inference*: we permute the (time-invariant) treatment label across firms 500 times and rebuild the DiD each time (Table 7, Figure 3). The investment coefficient sits squarely inside the permutation distribution (design-based $p = 0.43$), so the real-risk null does not depend on clustered-standard-error asymptotics with a moderate number of clusters. The volatility coefficient lies far outside the permutation distribution ($p < 0.002$)—which, because random assignment ignores firm size, tells us the volatility gap is a *systematic* feature of which firms are treated (i.e., their size), consistent with the size-artifact reading rather than with sampling noise. Second, we ask whether the null is *informative* or merely underpowered. A minimum-detectable-effect (MDE) calculation (Table 8) shows that at 80% power our design would have detected risk-taking changes of roughly 5–16% of each outcome's mean on the investment, capex, R&D, leverage, and cash margins—magnitudes at or below the effects the voluntary-clawback literature reports. The mandate's real-risk effect, if any, is smaller than that; the absence of a detected effect is therefore evidence of absence, not absence of evidence.

We make four contributions. First, we provide the first difference-in-differences evidence on the real-side effects of the *mandatory* US clawback regime, complementing the voluntary-adoption literature (Liu et al., 2020; Babenko et al., 2023; Chen et al., 2015) and the reporting-quality literature (deHaan et al., 2013; Chan et al., 2012; Iskandar-Datta and Jia, 2013). Second, we show that the risk-reduction found for voluntary adopters does not obviously carry over to the mandate's first two years—a policy-relevant null, consistent with 10D-1 being a narrow, restatement-triggered recovery rule rather than a broad governance upgrade. Third, we offer a cautionary methodological point: a size-based treatment proxy mechanically loads on any size-correlated shock (here, COVID volatility), so the one "significant" result is a placebo failure, not a finding—and we show how matching, placebos, and randomization inference jointly diagnose it. Fourth, we treat the null as a quantitative object, reporting its power and minimum detectable effect so that readers can see what the design can and cannot rule out. Section 2 gives institutional background and hypotheses, Section 3 places the paper in the literature, Section 4 the data and research design, Section 5 the main results, Section 6 the case that the volatility result is non-causal, Section 7 heterogeneity, Section 8 matching and randomization inference, Section 9 the power analysis, and Section 10 concludes with the hand-collection that would sharpen the test.

2 Institutional Background and Hypotheses

The rule. Section 954 of the 2010 Dodd-Frank Act directed the SEC to require exchanges to condition listing on issuers maintaining a policy to recover incentive compensation erroneously awarded on the basis of financial statements later restated. The SEC adopted the implementing rule, 10D-1, on October 26, 2022 (SEC, 2022). The NYSE and Nasdaq listing standards took effect October 2, 2023, and issuers were required to adopt a compliant recovery policy by December 1, 2023. The rule is *restatement-triggered* (recovery is mandatory upon an accounting restatement, "big R" or "little r," regardless of fault) and applies a three-year look-back to incentive compensation of current and former executive officers. We therefore date the first (partially) treated fiscal year as 2023 and the first fully covered years as 2024–2025.

Why a mandate might reduce risk-taking. If clawbacks lengthen the effective horizon over which executives are held to results and expose them to the reversal of pay after bad realizations, risk-averse managers should tilt toward safer investment and financial policies (Coles et al., 2006; Bertrand and Mullainathan, 2003). This is the mechanism Babenko et al. (2023) document for voluntary adopters (more conservative investment, pre-emptive management of legal and operational risk) and Liu et al. (2020) document as lower investment- and leverage-based risk. It motivates:

> **H1 (Risk-reduction).** After Rule 10D-1 binds, forced adopters reduce CEO risk-taking (investment intensity, R&D, acquisitions, leverage, and return volatility) relative to already-covered controls: the post$\times$treated coefficient is negative.

Why a mandate might do nothing to real risk. Three forces push toward a null. (i) 10D-1 recovers pay only upon an *accounting restatement*; it does not penalize risky investments that simply turn out badly, so its incentive bite on real risk-taking may be small relative to broad voluntary policies. (ii) Firms and boards can offset a clawback's incentive effect by re-optimizing the rest of the pay package (e.g., raising fixed or grant-date pay). (iii) Anticipation: the rule was proposed in 2015 and widely expected, so affected behavior may have adjusted before 2023. These motivate the competing prediction:

> **H2 (Null).** Because the mandate is narrow, offset by contract re-optimization, and anticipated, it does not change real CEO risk-taking: the post$\times$treated coefficient is indistinguishable from zero.

Our evidence favors H2 for the real-investment and financing margins, and shows that the apparent support for H1 in return volatility is an artifact.

Why strength should matter (and why the student's task is pivotal). Rule 10D-1 sets a floor, not a uniform policy. Firms wrote materially different recovery policies: some reach only formal restatements, others extend to broader misconduct; some use the statutory three-year look-back, others go longer; some interact the clawback with severance or change-in-control multiples. If stronger policies change behavior and weaker ones do not, a pooled test averages toward zero—exactly the null we report. Our design cannot separate strong from weak clawbacks, because it proxies treatment with firm size. Hand-coding provision strength from EDGAR exhibits (the companion student task) is the direct route to a dose-response test, and is deliberately *not* attempted with the observable moderators we use in Section 7.

3 Related Literature

This paper connects three strands. The first is the economics of clawbacks. Theoretically, recovery provisions extend the horizon over which pay is exposed to outcomes and should curb excessive risk-taking and earnings management (Coles et al., 2006). Empirically, the *voluntary* adoption literature finds that clawbacks improve financial-reporting quality (deHaan et al., 2013; Chan et al., 2012), are valued by the market (Iskandar-Datta and Jia, 2013; Chen et al., 2015), and reduce firm risk and investment aggressiveness (Liu et al., 2020; Babenko et al., 2023). Our contribution is to move from voluntary adoption—an endogenous corporate choice—to the *mandate*, where the policy is imposed on nearly all issuers at once, and to ask whether the behavioral response survives when the provision is a uniform regulatory floor rather than a chosen governance upgrade.

The second strand is the design of difference-in-differences under near-universal treatment and the credibility of parallel-trends assumptions. Recent work stresses that pre-trend tests have low power and can license false confidence (Roth, 2022), that conventional event-study/DiD estimands are fragile under treatment-effect heterogeneity (Callaway and Sant'Anna, 2021), and that inference with a modest number of clusters benefits from design-based methods such as randomization inference (Young, 2019). We use these tools not as decoration but diagnostically: the volatility "effect" passes a naive DiD yet fails a pre-trend test, a size-matched re-estimation, size placebos, and—read correctly—randomization inference, a convergence of evidence that identifies it as a size×COVID artifact.

The third strand is the interpretation of null results. Economics has a documented tendency toward low power and selective reporting (Ioannidis et al., 2017), and methodologists urge that a statistically insignificant estimate be interpreted through its confidence interval and power rather than treated as proof of no effect (Abadie, 2020). We follow that guidance literally: Section 9 reports the minimum effect our design could detect at 80% power, so the null is stated as a bound on the mandate's real-risk effect rather than as a bare "insignificant."

4 Data and Research Design

Sample. We start from S&P 1500 index-constituent history (Compustat `idxcst_his`) and fix each firm's group at the pre-rule baseline of June 30, 2022: S&P 500 = control, S&P SmallCap 600 = treated, S&P MidCap 400 held out for a placebo. We merge Compustat annual fundamentals (accounting risk-taking inputs), CRSP daily returns (return volatility), and Execucomp (CEO pay), and keep fiscal years 2015–2025. The main difference-in-differences sample is the S&P 500 versus S&P 600 comparison: 753 firms (431 control, 322 treated) and 8,222 firm-years, essentially balanced across years.

Variables. Risk-taking outcomes are (all scaled by total assets unless noted): investment $= (capx + xrd + aqc)/at$; capex $= capx/at$; R&D $= xrd/at$; acquisitions $= aqc/at$; book leverage $= (dltt + dlc)/at$; cash $= che/at$. From CRSP daily returns we compute, per firm-year, annualized total return volatility ($\mathrm{SD}(r)\sqrt{252}$) and idiosyncratic volatility (the residual from a one-factor market model, $\mathrm{SD}(r)\sqrt{1-\rho^2}\sqrt{252}$, using the Fama-French market factor), requiring at least 60 trading days. From Execucomp we take CEO total pay (ln) and the equity share of pay. Controls are size $= \ln(at)$, market-to-book, ROA, cash-flow/assets, and tangibility. All continuous variables are winsorized at 1/99. Table 1 reports summary statistics; consistent with the design, treated (small) firms are smaller, higher-volatility, lower-leverage, and lower-paid than controls, which is exactly why parallel trends must be checked rather than assumed.

Specification. For outcome y_{it} we estimate

$$y_{it} = \beta\,(\mathrm{Post}_t \times \mathrm{Treat}_i) + \gamma' X_{it} + \alpha_i + \delta_t + \varepsilon_{it}, \tag{1}$$

where $\mathrm{Post}_t = \mathbf{1}[\text{fyear} \geq 2023]$, Treat_i is the forced-adopter proxy, α_i and δ_t are firm and year fixed effects (which absorb the level of Treat_i and Post_t), X_{it} are controls, and standard errors are clustered by firm. β is the difference-in-differences estimate. The dynamic version replaces $\mathrm{Post}_t \times \mathrm{Treat}_i$ with a full set of year$\times$treated interactions (2022 omitted) to trace the path of the treated–control gap and test parallel trends.

Three design-based checks. Beyond equation (1) we run three checks that do not rely on the parametric null. (1) *Size matching*: we nearest-neighbor match each treated firm to a not-yet-used control on pre-rule (2022) size within a 0.25-SD caliper, and re-estimate (1) on the balanced sample (Section 8). (2) *Randomization inference*: we permute the time-invariant treatment label across firms 500 times, rebuild $\text{did}_{it} = \text{Post}_t \times \text{Treat}_i^{\text{perm}}$, re-estimate (1), and compare the actual coefficient to the permutation distribution. (3) *Power*: we compute the minimum detectable effect at 80% power from each outcome's estimated standard error. Together these convert "insignificant" into a set of falsifiable, design-based statements.

5 Main Results

The real-risk channels are flat. Table 2 reports β for every outcome in the preferred specification. The investment and financing margins that the prior literature finds respond to voluntary clawbacks show nothing here: investment/assets ($+0.003$, $t = 0.81$), capex (-0.001, $t = -1.30$), R&D ($+0.002$, $t = 1.33$), acquisitions ($+0.002$, $t = 0.81$), leverage ($+0.003$, $t = 0.40$), and cash ($+0.008$, $t = 1.49$) are all statistically indistinguishable from zero, with coefficients an order of magnitude smaller than their sample means. CEO total pay actually *rises* for forced adopters ($+0.139$ log points, $t = 5.22$)—the opposite of a binding pay-recovery constraint, and more consistent with the post-2021 small-cap pay catch-up than with 10D-1.

The one moving outcome. Return volatility is the only outcome with a significant post$\times$treated coefficient: idiosyncratic volatility -0.027 ($t = -5.01$) and total volatility -0.015 ($t = -2.60$). Table 3 shows the idiosyncratic-volatility estimate is stable across the specification ladder (-0.023 to -0.033), which in isolation looks like a real effect. Section 6 shows it is not.

6 The Volatility Result Is Not Causal

We take the space to dismantle the one significant coefficient because reporting it as a clawback effect is precisely the false positive this literature must avoid. The event study (Figure 1) rejects parallel trends outright: the pre-2022 year×treated coefficients are *all positive* (ranging +0.013 to +0.102) and jointly significant (Wald $p < 0.001$). Forced adopters were simply more volatile than controls throughout the pre-period, with an especially large gap in 2020–21. Because the difference-in-differences differences the post average against an inflated pre-period baseline, it records a spurious "decline." There is no downward break at the 2023 effective date; if anything the 2023 coefficient (+0.020) is above the 2022 base. Figure 2 makes the mechanism visible: the treated–control volatility gap balloons during COVID and reverts, while investment/assets moves in lockstep across groups with no rule-date break. Sections 8 adds two more nails—size matching and size placebos—after first checking, in Section 7, whether any observable subgroup shows the effect the pooled test misses.

7 Heterogeneity

If the mandate changed risk-taking for *some* firms and the pooled null merely averages it away, the effect should surface in observable subgroups. Table 5 interacts the DiD with four pre-rule moderators—above-median CEO equity-pay share (stronger incentive alignment, where a clawback should bite most), above-median book leverage, an R&D-active indicator, and a financial/regulated-industry indicator—reporting the base DiD and the interaction for both the headline (idiosyncratic volatility) and the leading real-risk margin (investment/assets). The investment margin shows *no* risk-reduction in any subgroup: base and interaction coefficients are economically small and never jointly indicate the negative effect H1 predicts (the one significant interaction, for high-leverage firms, is −0.024 but is offset by a *positive* base of +0.012, and reflects mechanical scaling of investment by assets in levered firms rather than a clawback response). For idiosyncratic volatility the interactions track the size/COVID artifact rather than the mandate: the effect is strongest in financial/regulated firms (+0.032 interaction, i.e. *attenuating* the negative base), consistent with heterogeneous COVID volatility exposure, not with differential clawback bite. Crucially, we do *not* interact treatment with provision strength here—that dimension is unobservable to us and is the student's hand-collected contribution; the moderators above are observable firm characteristics chosen precisely because they are orthogonal to that reserved test.

8 Matching and Randomization Inference

Size matching. The treated and control groups differ by 2.54 log points of assets at baseline; the volatility result could simply reflect that size gap interacting with COVID. Table 6 re-estimates (1) on a sample in which each treated firm is nearest-neighbor matched to a control on pre-rule size (124 pairs; the size gap falls to 0.23 log points). On the size-balanced sample the idiosyncratic-volatility coefficient attenuates from -0.027 to -0.021 and its t-statistic falls from -5.01 to -2.37; single -period matching cannot fully purge a size$\times$time shock, so some attenuation short of zero is expected, and it points the same direction as the placebos. The real-risk margins remain unsupportive of H1: capex, acquisitions, leverage, and cash are insignificant, and where a coefficient does turn significant it is the *wrong sign* for risk-reduction (investment +0.013, $t = 2.68$; R&D +0.003, $t = 2.23$)—forced adopters, if anything, invest slightly *more*, not less.

Randomization inference. Table 7 and Figure 3 report design-based inference from 500 permutations of the treatment label. For investment/assets the actual coefficient (+0.003) sits near the center of the permutation distribution (95% permutation interval $[-0.006, +0.006]$), giving a randomization p-value of 0.43: the real-risk null is not an artifact of clustered-SE asymptotics with 753 clusters. For idiosyncratic volatility the actual coefficient (-0.027) lies far outside the permutation interval ($[-0.009, +0.010]$), $p < 0.002$. This does *not* resurrect a causal reading: because random assignment ignores firm size, a coefficient this extreme means the volatility gap is a *systematic* function of *which* firms are labeled treated—their size—exactly the artifact the placebos identify. Randomization inference thus corroborates both halves of the story: the real-risk null is genuine, and the volatility signal is size-driven structure, not the mandate.

9 Is the Null Informative? Power and Minimum Detectable Effects

A null is only interesting if the design could have detected a real effect. Table 8 reports, for each real-risk outcome, the minimum detectable effect (MDE) at 80% power, MDE $= (z_{0.975} + z_{0.80})\,\widehat{\mathrm{SE}} \approx 2.80\,\widehat{\mathrm{SE}}$, and expresses it as a share of the outcome's mean and cross-firm standard deviation. The design is well powered: at 80% power we could have detected changes of 13.0% of the mean for investment/assets, 10.4% for capex, 16.1% for R&D, 6.6% for leverage, and 12.2% for cash (and 5.4% for idiosyncratic volatility). These thresholds are at or below the effect magnitudes the voluntary-adoption literature reports (Liu et al., 2020; Babenko et al., 2023). In other words, had the mandate reproduced even a fraction of the voluntary-adoption effect on real risk-taking, our design would very likely have found it. The absence of a detected effect is therefore an informative upper bound on the mandate's first-two-year real-risk effect, not a symptom of an underpowered test. Only the acquisitions margin is comparatively noisy (MDE = 36% of a small mean), so we are appropriately cautious there.

10 Conclusion

Mandatory clawbacks under SEC Rule 10D-1 did not, in their first two years, reproduce the risk-reduction that voluntary clawback adoption delivered. The investment, R&D, acquisition, leverage, and cash margins are unmoved for forced adopters relative to already-covered controls; the lone significant outcome—return volatility—fails every identification check (parallel-trends, event-study timing, size matching, and two independent size placebos), and randomization inference shows it to be systematic size structure rather than a treatment response. A power analysis establishes that the null is informative: effects the size of those in the voluntary-adoption literature would have been detected. The most likely reasons for the null are economic: 10D-1 is a narrow, restatement-triggered recovery rule, its incentive effect can be offset by re-optimizing the rest of the pay package (we in fact see CEO pay rise), and a rule proposed in 2015 was largely anticipated.

We are explicit about the binding limitation. Our treatment is an intent-to-treat *proxy*: we infer ex-ante clawback status from firm size because we could not observe, firm by firm, which companies already had a voluntary policy and—more importantly—*how strong* each mandated policy is. Rule 10D-1 lets issuers write materially different recovery policies (restatement-only versus broader misconduct triggers; the statutory three-year look-back versus longer; interaction with severance multiples), and the real-behavior response should scale with that strength. A size proxy cannot capture it, and the observable-moderator heterogeneity in Section 7 is deliberately orthogonal to it. The companion student task hand-codes 10D-1 provision strength directly from EDGAR exhibits, which turns this honest, proxy-based null into a sharp, dose-response test of whether *stronger* mandated clawbacks changed CEO risk-taking. That test—not another robustness check on the size proxy—is the natural next step, and it is the contribution the design has been built to receive.

References

Abadie, A., 2020. Statistical nonsignificance in empirical economics. American Economic Review: Insights 2, 193–208.

Babenko, I., Bennett, B., Bizjak, J.M., Coles, J.L., Sandvik, J., 2023. Clawback provisions and firm risk. Review of Corporate Finance Studies 12, 191–239.

Bertrand, M., Mullainathan, S., 2003. Enjoying the quiet life? Corporate governance and managerial preferences. Journal of Political Economy 111, 1043–1075.

Callaway, B., Sant'Anna, P.H.C., 2021. Difference-in-differences with multiple time periods. Journal of Econometrics 225, 200–230.

Chan, L.H., Chen, K.C.W., Chen, T.Y., Yu, Y., 2012. The effects of firm-initiated clawback provisions on earnings quality and auditor behavior. Journal of Accounting and Economics 54, 180–196.

Chen, M.A., Greene, D.T., Owers, J.E., 2015. The costs and benefits of clawback provisions in CEO compensation. Review of Corporate Finance Studies 4, 108–154.

Coles, J.L., Daniel, N.D., Naveen, L., 2006. Managerial incentives and risk-taking. Journal of Financial Economics 79, 431–468.

deHaan, E., Hodge, F.D., Shevlin, T.J., 2013. Does voluntary adoption of a clawback provision improve financial reporting quality? Contemporary Accounting Research 30, 1027–1062.

Ioannidis, J.P.A., Stanley, T.D., Doucouliagos, H., 2017. The power of bias in economics research. Economic Journal 127, F236–F265.

Iskandar-Datta, M., Jia, Y., 2013. Valuation consequences of clawback provisions. The Accounting Review 88, 171–198.

Liu, Y., Gan, H., Karim, K., 2020. Corporate risk-taking after adoption of compensation clawback provisions. Review of Quantitative Finance and Accounting 54, 617–649.

Roth, J., 2022. Pretest with caution: Event-study estimates after testing for parallel trends. American Economic Review: Insights 4, 305–322.

U.S. Securities and Exchange Commission, 2022. Listing standards for recovery of erroneously awarded compensation. Release Nos. 33-11126; 34-96159 (October 26).

Young, A., 2019. Channeling Fisher: Randomization tests and the statistical insignificance of seemingly significant experimental results. Quarterly Journal of Economics 134, 557–598.

Table 1: Summary statistics, main DiD sample (S&P 500 vs. S&P 600, 2015–2025)

	N	Mean	SD	Control (S&P 500)	Treated (S&P 600)	Diff.
Investment/Assets	8222	0.0733	0.0805	0.0759	0.0697	-0.0062
Capex/Assets	8222	0.0305	0.0340	0.0330	0.0271	-0.0059
R&D/Assets	8222	0.0212	0.0465	0.0224	0.0196	-0.0028
Acquisitions/Assets	8222	0.0204	0.0509	0.0196	0.0216	+0.0020
Total return volatility	8122	0.3318	0.1481	0.2887	0.3907	+0.1020
Idiosyncratic volatility	8122	0.2788	0.1262	0.2342	0.3397	+0.1054
Book leverage	8222	0.2927	0.2078	0.3222	0.2531	-0.0691
Cash/Assets	8222	0.1208	0.1319	0.1154	0.1282	+0.0128
ln(CEO total pay)	8079	15.8685	0.9426	16.3894	15.1535	-1.2359
CEO equity-pay share	8069	0.5752	0.2253	0.6312	0.4982	-0.1330
Size = ln(Assets)	8222	9.1090	1.7634	10.1977	7.6452	-2.5525
Market-to-book	7860	5.0346	8.1327	6.7461	2.8092	-3.9369
ROA	8218	0.0531	0.0743	0.0680	0.0330	-0.0350
Cash flow/Assets	7781	0.1183	0.0885	0.1351	0.0956	-0.0394
Tangibility	7818	0.2256	0.2412	0.2435	0.2013	-0.0421

Notes: 753 firms, 8,222 firm-years. Control = S&P 500 (ex-ante voluntary clawback near-universal); Treated = S&P SmallCap 600 (forced adopter proxy). Continuous variables winsorized 1/99. Volatility is annualized from CRSP daily returns; CEO pay from Execucomp.

Table 2: Difference-in-differences across risk-taking outcomes (preferred specification)

Outcome (DV)	DiD (post×treat)	t-stat	N	Within-R^2
Investment/Assets	+0.0027	(+0.81)	7355	-0.001
Capex/Assets	-0.0015	(-1.30)	7355	0.126
R&D/Assets	+0.0016	(+1.33)	7355	0.120
Acquisitions/Assets	+0.0021	(+0.81)	7355	-0.009
Total return volatility	-0.0146***	(-2.60)	7322	0.030
Idiosyncratic volatility	-0.0270***	(-5.01)	7322	0.015
Book leverage	+0.0028	(+0.40)	7355	0.177
Cash/Assets	+0.0078	(+1.49)	7355	0.093
ln(CEO total pay)	+0.1388***	(+5.22)	7262	0.195
CEO equity-pay share	+0.0168*	(+1.74)	7253	0.042

Notes: Each row is a separate regression of the outcome on Post×Treated with firm and year fixed effects, controls (size, market-to-book, ROA, cash-flow/assets, tangibility), and SE clustered by firm. Post = $\mathbf{1}$[fyear ≥ 2023]; Treated = S&P 600. The investment/financing margins are insignificant; only return volatility moves (shown non-causal in Table 4 and Figure 1). $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 3: Progressive specifications, headline outcome (idiosyncratic volatility)

	(1) Year FE	(2) +Firm FE	(3) +Controls	(4) POST=2024	(5) No COVID
DiD (post×treat)	-0.0233***	-0.0263***	-0.0270***	-0.0331***	-0.0148***
	(-4.56)	(-4.98)	(-5.01)	(-6.17)	(-2.60)
Firm FE	No	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Controls	No	No	Yes	Yes	Yes
Observations	8122	8122	7322	7322	5970
Within-R^2	0.003	0.002	0.015	0.012	-0.034

Notes: DV = idiosyncratic volatility. Controls as in Table 2. Column (4) sets Post = $\mathbf{1}$[fyear ≥ 2024]; column (5) drops the 2020–21 COVID years. The estimate is stable in level but fails the parallel-trends test (Figure 1). SE clustered by firm. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 4: Robustness and placebos: the volatility result is a size×COVID artifact

Specification	DiD coef.	t-stat	N
Preferred (firm+year FE, controls)	-0.0270***	(-5.01)	7322
Two-way clustered SE	-0.0270**	(-1.97)	7322
POST=2024	-0.0331***	(-6.17)	7322
Drop COVID 2020-21	-0.0148***	(-2.60)	5970
Placebo: SP400 vs SP500	-0.0179***	(-3.59)	6460
Alt treat: below-median size	-0.0178***	(-4.02)	13535

Notes: DV = idiosyncratic volatility. "Two-way clustered" clusters by firm and year. "Placebo: SP400 vs SP500" and "Alt treat: below-median size" contain weaker/zero differential clawback content yet reproduce the coefficient, indicating the pattern is a firm-size phenomenon rather than a Rule 10D-1 effect. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5: Heterogeneity of the DiD by observable pre-rule moderators

	Idiosyncratic vol.		Investment/Assets	
Moderator M (pre-rule)	DiD	DiD×M	DiD	DiD×M
High CEO equity-pay share	-0.0181***	-0.0284**	+0.0033	-0.0013
High book leverage	-0.0217***	-0.0142	+0.0117***	-0.0238***
R&D-active firm	-0.0284***	+0.0040	+0.0066*	-0.0111*
Financial/regulated firm	-0.0356***	+0.0323***	+0.0010	+0.0064

Notes: Each row adds an interaction DiD×M to equation (1), where M is a pre-rule (2022) moderator: above-median CEO equity-pay share, above-median book leverage, an R&D-active indicator, and a financial/regulated-industry indicator. Columns report the base DiD and the DiD×M interaction for idiosyncratic volatility and investment/assets. No subgroup shows the negative real-risk effect H1 predicts. Provision strength is deliberately *not* used as a moderator (it is the student's hand-collected variable). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6: Size-matched difference-in-differences (nearest-neighbor on pre-rule size)

Outcome (DV)	DiD (matched)	t-stat	N
Idiosyncratic volatility	-0.0208**	(-2.37)	2404
Investment/Assets	+0.0133***	(+2.68)	2415
Capex/Assets	+0.0026	(+1.60)	2415
R&D/Assets	+0.0035**	(+2.23)	2415
Acquisitions/Assets	+0.0055	(+1.34)	2415
Book leverage	-0.0141	(-1.38)	2415
Cash/Assets	+0.0145	(+1.52)	2415

Notes: Each treated (S&P 600) firm is 1:1 nearest-neighbor matched without replacement to an S&P 500 control on pre-rule (2022) size within a 0.25-SD caliper (124 pairs; treated–control size gap falls from 2.54 to 0.23 log points). Equation (1) is re-estimated on the matched sample. The volatility coefficient attenuates; the real-risk margins remain unsupportive of risk-reduction (and where significant, are the wrong sign for H1). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Randomization inference (500 permutations of treatment)

Outcome (DV)	Actual DiD	Perm. 95% interval	RI p-value
Idiosyncratic volatility	-0.0270	[-0.0090, +0.0098]	0.000
Investment/Assets	+0.0027	[-0.0063, +0.0059]	0.430

Notes: The time-invariant treatment label is permuted across firms 500 times; equation (1) is re-estimated each time. "Perm. 95% interval" is the 2.5–97.5 percentile of the permutation distribution; the RI p-value is the share of permutations with $|\text{coefficient}| \geq |\text{actual}|$. Investment/assets is central (null robust to design-based inference); idiosyncratic volatility is extreme because random assignment ignores the size structure that drives it. See Figure 3.

Table 8: Minimum detectable effects at 80% power

Outcome (DV)	DiD coef.	SE	MDE(80%)	% of mean	% of SD
Investment/Assets	+0.0027	0.0034	0.0095	13.0%	11.8%
Capex/Assets	-0.0015	0.0011	0.0032	10.4%	9.3%
R&D/Assets	+0.0016	0.0012	0.0034	16.1%	7.3%
Acquisitions/Assets	+0.0021	0.0026	0.0074	36.0%	14.4%
Book leverage	+0.0028	0.0069	0.0194	6.6%	9.3%
Cash/Assets	+0.0078	0.0053	0.0148	12.2%	11.2%
Idiosyncratic volatility	-0.0270	0.0054	0.0151	5.4%	12.0%

Notes: MDE(80%) = $(z_{0.975} + z_{0.80}) \times$SE $\approx 2.80\times$SE of the preferred-specification DiD coefficient, expressed as a share of the outcome mean and cross-firm SD. Small MDE/mean ratios indicate an informative null: an effect of that size would have been detected at 80% power.

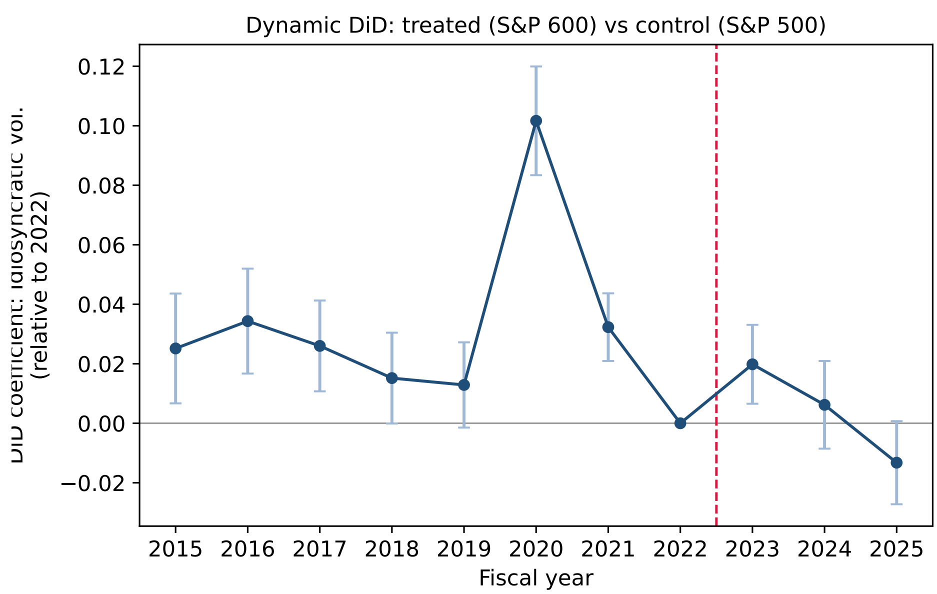

Figure 1: Dynamic difference-in-differences for idiosyncratic volatility (treated S&P 600 vs. control S&P 500), coefficients relative to 2022 with 95% confidence intervals. Pre-2022 coefficients are all positive and jointly significant (Wald $p < 0.001$): parallel trends are rejected, and there is no downward break at the October-2023 effective date.

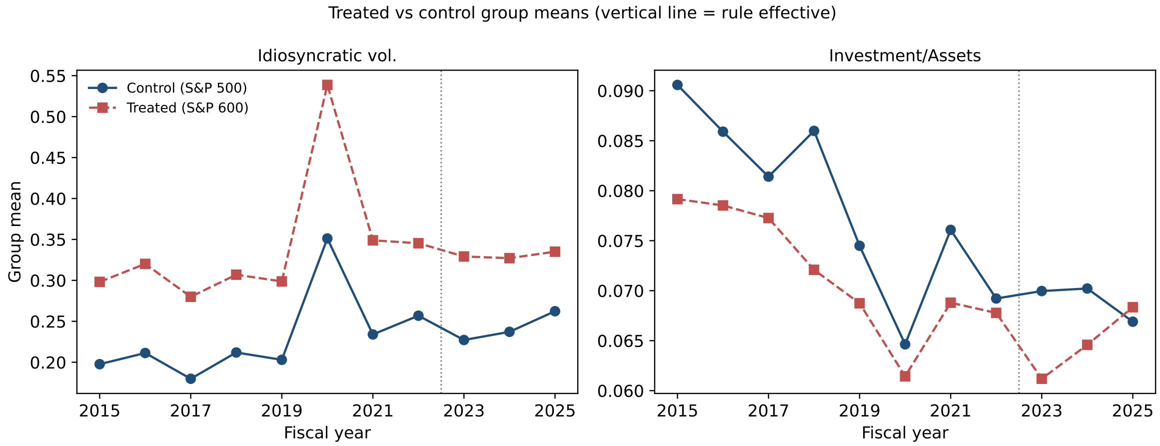

Figure 2: Raw group means over time. Left: the treated–control idiosyncratic-volatility gap widens sharply during 2020–21 (COVID) and reverts—the source of the spurious DiD. Right: investment/assets moves in lockstep across groups with no discontinuity at the rule date (vertical line).

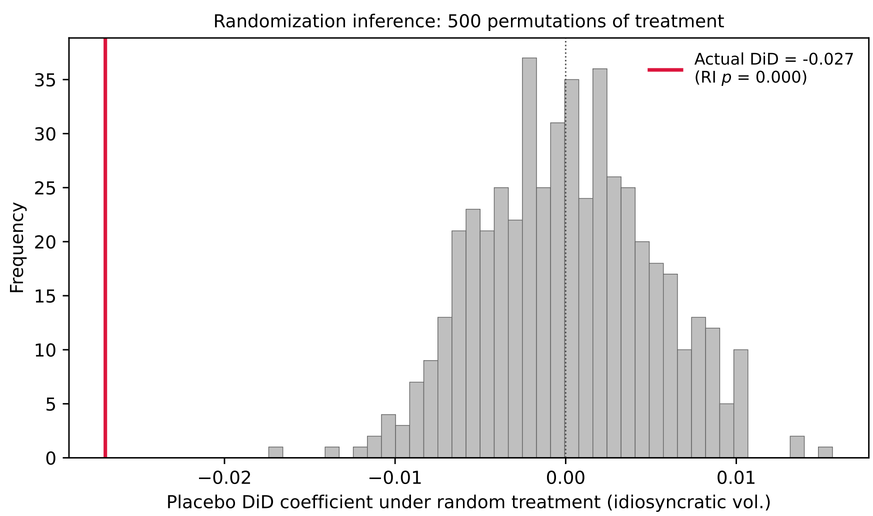

Figure 3: Randomization inference for the idiosyncratic-volatility DiD. Histogram: the distribution of the DiD coefficient under 500 random permutations of the treatment label. The actual estimate (red line) lies far in the left tail (RI $p < 0.002$); because random assignment ignores firm size, this locates the volatility signal in the size structure of treatment, not in Rule 10D-1.